

Cost Analysis

XBandDigitalModule_20x20 - Cost Analysis

Document Information

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1. Cost Summary

1.1 Unit Cost (Primary Configuration)

Category	Cost (USD)	% of Total
ADC Section	\$850	0.45%
FPGA Section	\$180,000	95.14%
Clock Section	\$450	0.24%
Power Section	\$334	0.18%
Passive Compone	\$100	0.05%
Mechanical	\$2,600	1.37%
Assembly	\$5,000	2.64%
Subtotal (Mod **\$189,334 **100%**		

1.2 Unit Cost (Alternative Configuration)

Category	Cost (USD)	Savings
ADC Section	\$650	\$200
FPGA Section	\$100,000	\$80,000
Clock Section	\$350	\$100
Power Section	\$300	\$34
Passive Compone	\$90	\$10
Mechanical	\$2,500	\$100
Assembly	\$5,000	\$0
Subtotal (Mod **\$108,890 **\$80,444**		

2. Cost Drivers

2.1 Primary Cost Driver: FPGA

The XQRCV1902 FPGA accounts for 95% of the module cost. This is due to:

2.2 Secondary Cost Drivers

Driver	Impact	Mitigation
Low volume	High unit cost	Volume discount
Long lead time	Inventory costs	Early procureme
Radiation testi	Qualification c	Shared testing

3. Volume Pricing

3.1 Quantity Breaks

Quantity	Unit Price	Total	Discount
1	\$189,334	\$189,334	0%
2-4	\$175,000	\$350,000-\$700,0	7.6%
5-9	\$165,000	\$825,000-\$1,485	12.8%
10-24	\$145,000	\$1,450,000-\$3,4	23.4%
25+	\$125,000	\$3,125,000+	34.0%

3.2 Alternative Configuration Pricing

Quantity	Unit Price	Total	Savings vs Prim
1	\$108,890	\$108,890	\$80,444 (42.5%)
5	\$95,000	\$475,000	\$425,000 (47.2%)
10	\$85,000	\$850,000	\$600,000 (41.4%)

4. Non-Recurring Engineering (NRE)

4.1 NRE Breakdown

Item	Cost (USD)	Notes
PCB Fabrication	\$5,000	Prototype run
SMT Stencil	\$500	For assembly
Test Fixtures	\$10,000	Custom test jig
Environmental T	\$25,000	Thermal, vibrat
Radiation Testi	\$25,000	TID, SEL, SEU
Documentation	\$5,000	Technical manua
Project Managem	\$10,000	Coordination
Total NRE	**\$80,500**	-

4.2 NRE Amortization

Production Volu	NRE per Unit	Total Unit Cost
1 unit	\$80,500	\$269,834
5 units	\$16,100	\$205,434
10 units	\$8,050	\$197,384
25 units	\$3,220	\$192,554

5. Life Cycle Cost

5.1 10-Year Mission Cost (10 Units)

Category	Cost (USD)
Module Cost (10	\$1,450,000
NRE	\$80,500
Integration	\$50,000
Testing	\$25,000
Spares (2 units	\$290,000
Total Life Cy	**\$1,895,500

5.2 Cost per Channel

Metric	Value
Total Cost	\$189,334
DDC Channels	8
Cost per Channe	\$23,667
Bandwidth per C	500 MHz
Cost per MHz	\$47.33

6. Cost Optimization Opportunities

6.1 Near-Term (0-6 months)

Opportunity	Potential Savin	Risk
Use ADC12DJ5200	\$200	Lower TID margi
Negotiate volum	5-10%	None
Optimize PCB la	\$500	Design complexi

6.2 Long-Term (6-12 months)

Opportunity	Potential Savin	Risk
Use XQRKU060 FP	\$80,000	Speed limitatio
Design for prod	10-15%	Redesign effort
Alternative sup	5-10%	Qualification

7. Budget Approval

Item	Budget	Actual	Status
Module (per uni	\$200,000	\$189,334	Under budget
NRE	\$100,000	\$80,500	Under budget
Testing	\$50,000	\$50,000	On budget
Total	**\$350,000**	**\$319,834**	**8.6% under**

8. Revision History

Version	Date	Author	Description
1.0	2026-09-09	Financial Analy	Initial release
